Artificial Intelligence Basics

Artificial intelligence helps computers recognize images, understand language, and make decisions.

Machine learning discovers patterns from training data. Good data improves predictions.

Supervised learning uses labeled examples. Classification predicts categories; regression predicts numbers.

Unsupervised learning discovers patterns without labels. Clustering groups similar examples.

Training sets teach parameters. Test sets evaluate unseen examples and must not be used in training.

Overfitting means a model performs well on training data but poorly on new examples.

Accuracy is correct predictions divided by all predictions. 85 correct predictions out of 100 equals 85 percent accuracy.

Deep learning uses multiple layers of neural networks.

Language models predict text from context. Prompts describe the task, context, and output format.

Context augmentation supplies relevant source material. Models can hallucinate, so important facts must be checked.